VendorPulse: Hyper-Local Early Warning for Informal Vendors

Executive Summary · ModelX 2.0 Competition Delivery

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1. Problem Statement: The Informal Vendor Predatory Debt Cycle

In dense urban commercial nodes such as Colombo 01–15 (Pettah Central, Fort, Maradana, Maligawatta), tens of thousands of informal hawkers and street vendors survive entirely on daily cash flow: **today's earnings directly fund tomorrow's inventory**. When acute localized shocks strike—such as sudden monsoon flash flooding, unannounced bus transit strikes (e.g., Route 138), or municipal road barricades—street-level foot traffic collapses immediately. Lacking institutional credit or personal savings, vendors face an instant liquidity crisis. To feed their families and preserve stall inventory, they are forced to borrow from unregulated predatory lenders charging ruinous interest rates of **10% to 20% weekly**, wiping out months of financial progress in a single weekend.

2. The Target User & The Paradigm Shift

VendorPulse is built specifically for a **Micro-Finance NGO Field Outreach Coordinator**. Currently, NGOs and micro-finance institutions operate retroactively: restructuring loans or processing relief weeks after defaults have occurred—when the vendor has already absorbed toxic micro-debt. **VendorPulse shifts the coordinator from reactive recovery to proactive shielding.** The coordinator consults the Threat Map 48–72 hours before Friday trading, sees a high-confidence collapse forecast, and dispatches rapid zero-interest bridge micro-loans to protect vendors before trading halts.

3. Core Innovation: Compounded Multi-Modal Disruption Extraction

Human operators cannot manually track and cross-reference thousands of shifting civic, transit, and meteorological signals across a city, nor quantify non-linear compounding effects. VendorPulse algorithmically extracts this intelligence by merging: (1) **OpenWeatherMap** 72-hour precipitation forecasts, (2) **OpenStreetMap (OSRM)** transit corridor telemetry, and (3) **spaCy-compatible NLP entity extraction** from local news RSS feeds (Ada Derana, NewsFirst), isolating commercial blocks like *Bodhiraja Mawatha* and *Bus Route 138*.

Key Paradigm Shift: Shifting resource allocation from lagging retroactive default restructuring to 48-to-72-hour preemptive zero-interest liquidity bridging that prevents informal daily-wage vendors from entering the predatory debt spiral.

4. End-to-End System Architecture

VendorPulse operates across four modular tiers: (1) **Data Ingestion** polling weather APIs, free OpenStreetMap transit telemetry, and civic scrapers every 3 hours; (2) **Processing & Storage** standardizing tensors to UTC+5:30 with Supabase PostgreSQL and SQLite local failover; (3) **Intelligence Layer** running a PyTorch Multi-Horizon Neural Forecaster with Quantile Loss (P_{50}, P_{90}); and (4) **Presentation Layer** providing a dark-mode mobile field interface with 1-click bridge capital authorization.

5. Empirical Benchmark Evaluation & Failure Criteria Verification

VendorPulse was empirically evaluated against a **600-day out-of-sample holdout dataset** of Colombo commercial disruptions, benchmarked directly against the two baselines defined in Section 9 of the proposal:

Model Architecture	Accuracy	Precision	Recall	F1-Score	FPR (< 20%)	P50 Loss
VendorPulse Multi-Horizon	**96.7%**	72.3%	82.9%	**0.773**	**2.3% (PASS)**	0.0359
Naïve Baseline ($T_{+1} = T_0$)	88.0%	12.2%	12.2%	0.122	6.4% (PASS)	0.0733
Traditional ARIMA (Statistical)	93.2%	0.0%	0.0%	0.000	0.0% (PASS)	0.0541

- **False Positive Rate < 20%: PASS (2.33%)** — Avoids coordinator alert fatigue and prevents capital wastage.
- **Significantly Outperformed Naïve Baseline: PASS (0.773 vs 0.122)** — Demonstrates real multi-modal predictive signal.
- **Outperformed Traditional ARIMA: PASS (0.773 vs 0.000)** — Confirms linear models fail on acute step-function shocks.

6. Field Interface & Concrete Social Impact

In the live interface, the coordinator views an interactive Threat Map and taps **Pettah Intersection (Block P-04)**, displaying: **90% Confidence**, **142 Registered Vendors**, **Rs 2.5M Capital Required**, and live signals (*Flooded primary drainage* and *Bus Route 138 suspended*). Tapping **'Authorize Bridge Deployment'** logs the decision to Supabase and immediately alerts field officers. By deploying Rs 2.5M in zero-interest bridge liquidity, 142 vendors save over **Rs 500,000 per week in predatory interest payments**, safeguarding informal families from poverty traps.

Public GitHub Repository: https://github.com/Sahas-sass/VendorPulse
Live Backend: FastAPI on port 8000 | **Frontend:** React 18 Dark-Mode Field Map on port 5173